Part 1: Theory

1. What is a Power Supply System?

A Power Supply System provides electrical power to a computer and its components.

Electricity from the wall socket cannot be used directly by computer components because:

- Wall power is AC (Alternating Current).
- Computer components need DC (Direct Current).

Therefore, a device is needed to convert AC to DC.

This device is called an **SMPS**.

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2. What is SMPS?

SMPS = Switched Mode Power Supply

It is a hardware component inside the computer cabinet that converts AC power into different DC voltages required by computer components.

Functions of SMPS

- Converts AC to DC.
- Supplies power to motherboard.
- Supplies power to CPU.

- Supplies power to hard disk/SSD.
- Supplies power to graphics card.
- Protects components from voltage fluctuations.

Common Output Voltages

Voltage	Usage
+12V	CPU, Fans, Graphics Card
+5V	USB Devices, Hard Disk
+3.3V	RAM, Motherboard Components

Real-Life Example

Think of SMPS as a translator.

Wall Socket → AC Power

Computer Components → Need DC Power

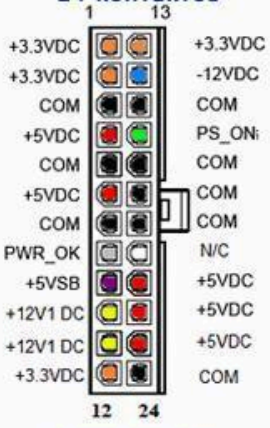
SMPS translates AC into DC so components can understand and use it.

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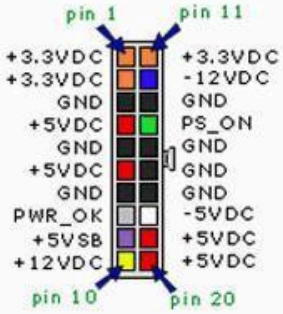
3. Parts of an SMPS



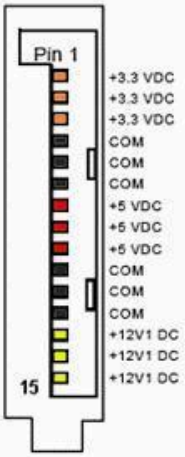
Главный разъем 24 контактов



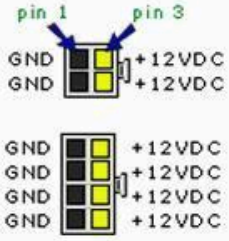
Главный разъем 20 контактов



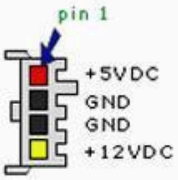
Разъем для подключения питания SATA жесткого диска



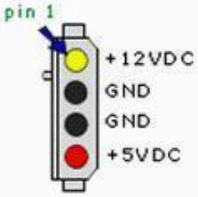
Разъемы для дополнительного питания процессора

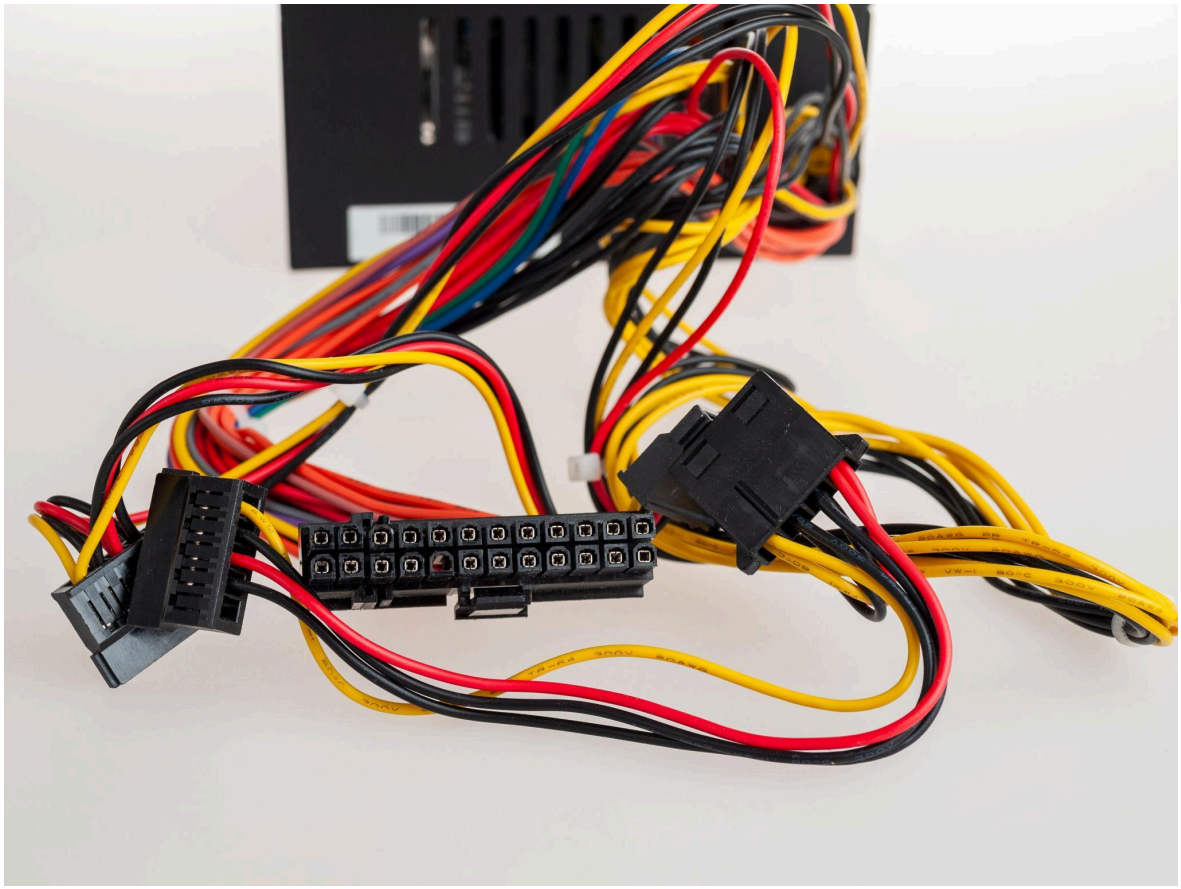


Разъем для подключения питания флопи дисковода



Разъем Molex для подключения питания различных устройств (жесткий диск, вентиляторы ..)







Main Parts

1. Cooling Fan
    - Prevents overheating.
2. Power Switch
    - Turns power supply ON/OFF.
3. Power Socket
    - Connects AC cable.
4. Output Connectors
    - Connect motherboard and storage devices.

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4. Advantages of SMPS

- High efficiency
- Less heat generation
- Compact size
- Lightweight
- Reliable operation

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5. What is UPS?

UPS = Uninterruptible Power Supply

A UPS provides backup power when electricity fails.

Functions of UPS

- Provides temporary backup power.
- Protects against power cuts.
- Protects against voltage fluctuations.
- Allows safe shutdown of computer.

Real-Life Example

Imagine you are saving an important project.

Suddenly electricity goes off.

Without UPS:

- Computer shuts down immediately.

- Unsaved work may be lost.

With UPS:

- Computer continues running for a few minutes.
- You can save your work safely.

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6. Types of UPS

1. Offline UPS

- Most common.
- Used in homes and small offices.

2. Online UPS

- Provides continuous power.
- Used in servers and data centers.

3. Line Interactive UPS

- Better voltage regulation.
- Used in offices.

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7. SMPS vs UPS

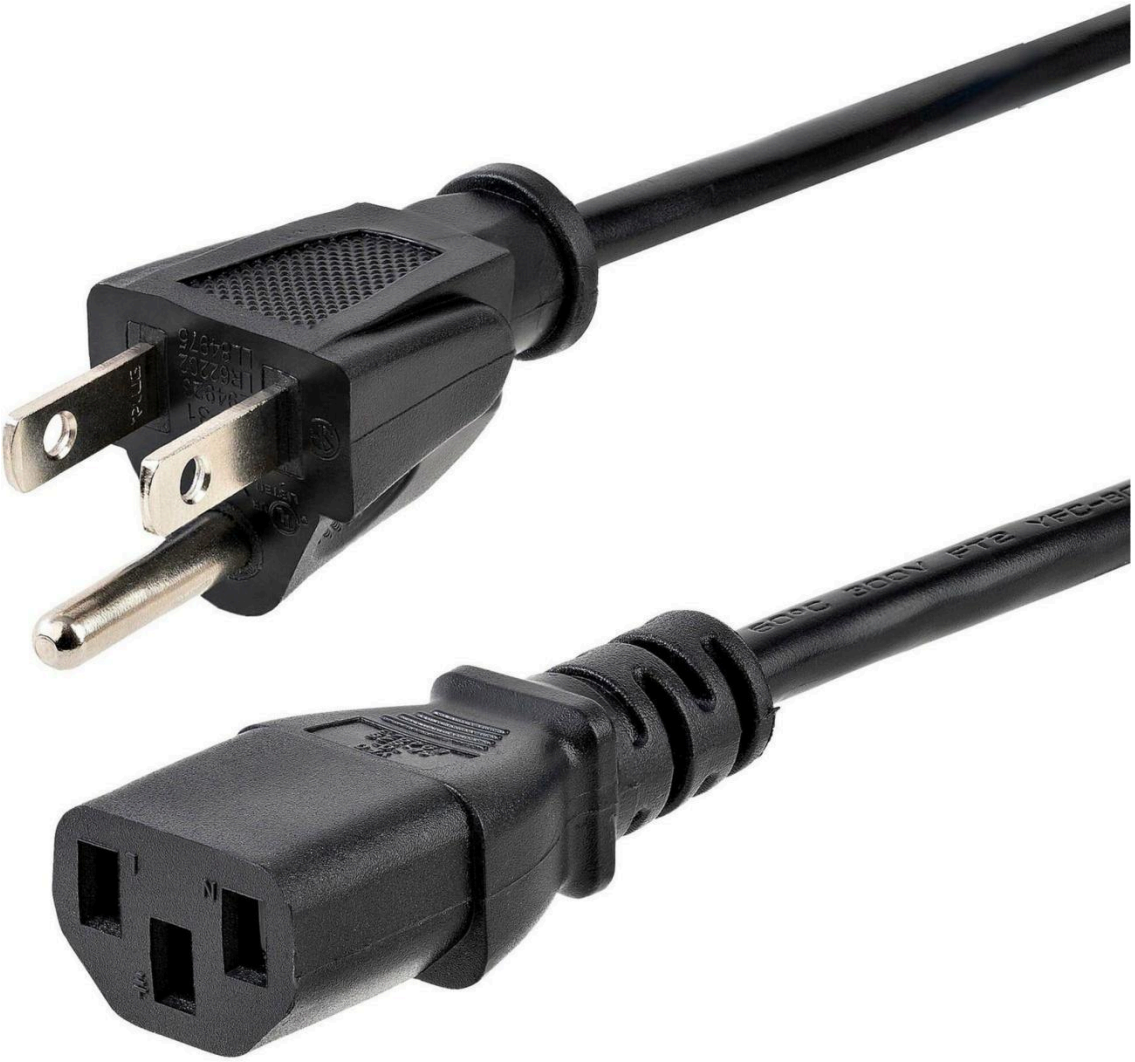
Feature	SMPS	UPS
Full Form	Switched Mode Power Supply	Uninterruptible Power Supply

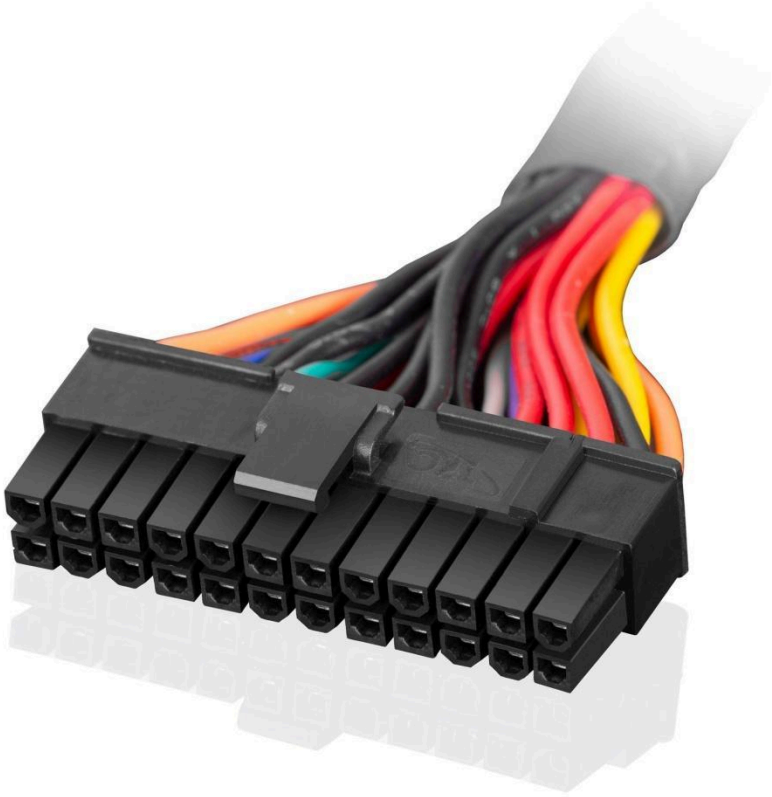
Feature	SMPS	UPS
Purpose	Converts AC to DC	Provides Backup Power
Installed	Inside Computer	Outside Computer
Backup Battery	No	Yes
Protects From	Voltage Issues	Power Failure

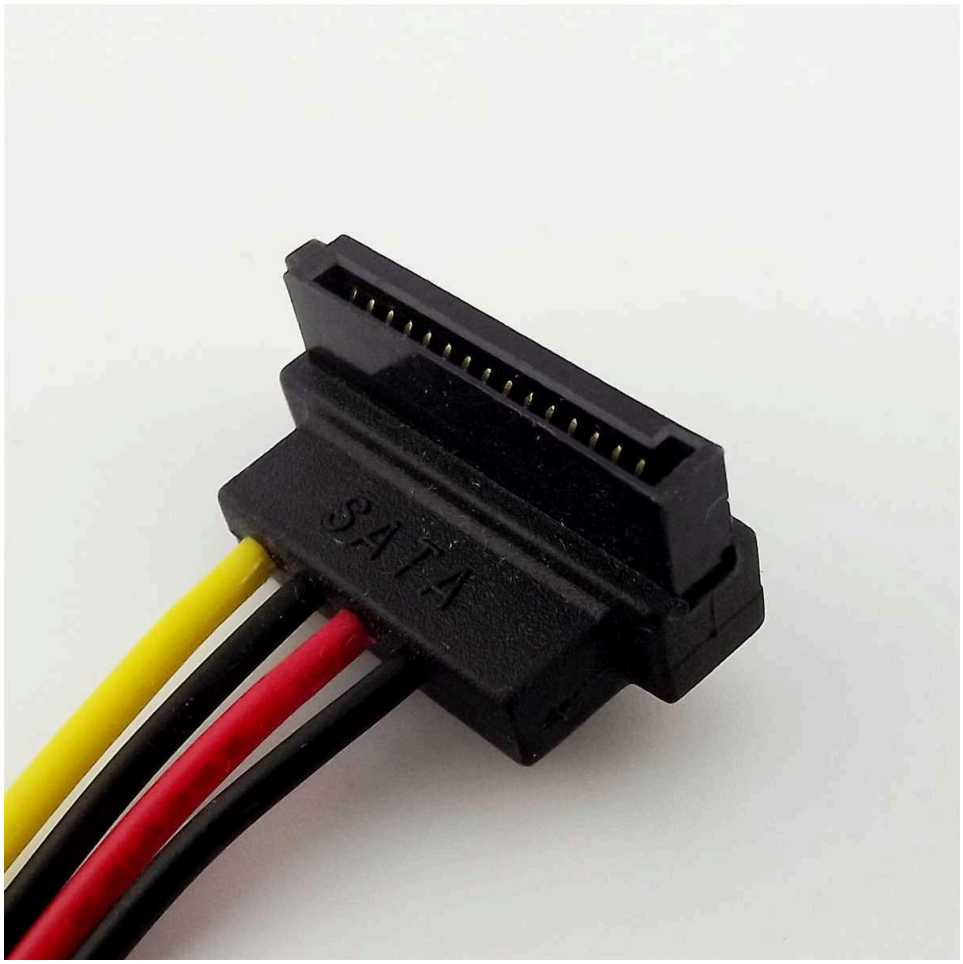
Part 2: Practical

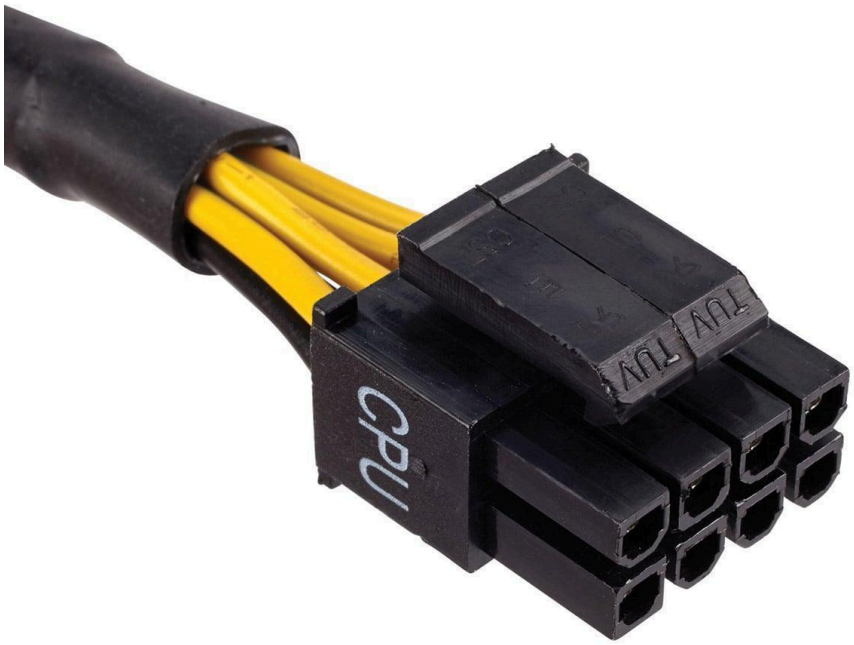
Practical 1: Identify Power Cables

Ask students to identify the following cables.






SATA


CPU
TUV
TUV



A. AC Power Cable

Used to connect:

- UPS to Wall Socket
- Computer to UPS

Identification:

- Thick cable
- Three-pin connector

B. 24-Pin ATX Connector

Used to provide power to:

- Motherboard

Identification:

- Largest connector from SMPS.

C. SATA Power Connector

Used for:

- SSD
- HDD
- DVD Drive

Identification:

- Flat black connector.

D. CPU Power Connector (4/8 Pin)

Used to power:

- Processor

Connected near CPU socket on motherboard.

Practical 2: Locate Power Components

Students should open a CPU cabinet and identify:

✓ SMPS

✓ Motherboard Power Connector

✓ CPU Power Connector

✓ SATA Power Connector

✓ Cooling Fan

✓ Power Switch

Practical 3: Check UPS Backup

Steps

1. Turn ON computer.
2. Connect through UPS.
3. Remove UPS input power.
4. Observe system remains ON.

Discussion:

- How long does UPS provide backup?
- Why is UPS important?

Practical 4: Check Battery Status Using CMD

Open Command Prompt:

```
powercfg /batteryreport
```

Generates battery report (for laptops).

Report location:

```
C:\Users\Username\battery-report.html
```

Students can open the report in browser.

Practical 5: View Power Configuration

```
powercfg /a
```

Shows available sleep and power states.

Practical 6: See System Uptime

```
systeminfo | find "Boot Time"
```

OR

```
net statistics workstation
```

Shows when the system was last started.

Practical 7: Shutdown Timer

```
shutdown /s /t 60
```

Computer will shut down after 60 seconds.

Cancel shutdown:

```
shutdown /a
```

Students usually find this practical interesting.

Practical 8: Restart Computer After 30 Seconds

```
shutdown /r /t 30
```

Explanation:

- `/r` = Restart
- `/t` = Time in seconds

Cancel:

```
shutdown /a
```

Find Biggest Files on PC

File Explorer Search:

```
size:>100MB
```

OR

```
size:>1GB
```

Students can identify storage-hungry files.

View Hidden Files

File Explorer → View → Hidden Items

Show:

- AppData folder
- Hidden system files

Explain why malware often hides here.

Open Notepad.

Type:

```
.LOG
```

Save file.

Close and reopen it.

Personal Message Box

Create a file:

```
message.vbs
```

Paste:

```
MsgBox "Welcome to Today's Practical Session!"
```

Save and double-click.

A popup message appears.

Students can customize their own messages.

Make the Computer Speak

Create:

```
speak.vbs
```

Paste:

```
Set speech=CreateObject("sapi.spvoice")
speech.Speak "Hello Students, Welcome
to Computer Lab"
```

Save and run.

Computer will speak the message.

Generate Random Numbers

CMD:

```
echo %random%
```

Run multiple times.

Create a simple guessing game.

Example:

```
echo Guess number between 1 and 100
echo Actual Number = %random%
```

Digital Clock in CMD

Create:.bat file

```
@echo off
:time
cls
echo %time%
```

```
timeout /t 1 >nul
goto time
```

Find All IP Addresses in Network

CMD:

```
arp -a
```

Shows devices recently connected on the network.

Students get excited seeing multiple IP addresses.

Funny Folder Names

Create folders named:

```
Homework
Exams
Secret Files
```

Then use:

```
attrib +h Homework
```

Folder disappears.

Show hidden items to bring it back.

Useful introduction to file attributes.

Folder Attributes Practical (Windows CMD)

Step 1: Create Folders

Open **Command Prompt (CMD)** and run:

```
mkdir Homework
mkdir Exams
mkdir "Secret Files"
```

Or create all folders at once:

```
mkdir Homework Exams "Secret Files"
```

Step 2: Hide the Homework Folder

```
attrib +h Homework
```

+h = Hidden Attribute

The **Homework** folder will no longer be visible in File Explorer.

Step 3: Show Hidden Items

1. Open **File Explorer**.
2. Click on the **View** menu.
3. Select **Show → Hidden Items**.

The **Homework** folder will become visible again.

Step 4: Unhide the Folder

Run the following command in CMD:

```
attrib -h Homework
```

-h = Remove Hidden Attribute

Extra Practical Activity

Make the folder **Hidden + System**:

```
attrib +h +s Homework
```

Now the folder may remain hidden even when **Hidden Items** is enabled.

To restore it:

```
attrib -h -s Homework
```

Check Folder Attributes

```
attrib Homework
```

Example Output:

```
SH      Homework
```

Attribute Meanings

Attribute	Meaning
S	System
H	Hidden
R	Read Only
A	Archive

Create Password Protected ZIP

Right-click folder → Send To → Compressed Folder.

Then use WinRAR/7-Zip to add password.

Teach basic file security.

Encrypt Drive

Encrypt pendrive

Encrypt document

Practical: Password Protect a Microsoft Word Document

Steps

1. Open **Microsoft Word**.

2. Create a new document and add some text.
3. Click **File** (or Office Button in Word 2007).
4. Select **Info**.
5. Click **Protect Document**.
6. Choose **Encrypt with Password**.
7. Enter a password.
8. Re-enter the password to confirm.
9. Save the document.

Test

1. Close the document.
2. Open it again.
3. Word will ask for the password before opening the file.

Remove Password

1. Open the document using the password.
2. Go to **File → Info → Protect Document → Encrypt with Password**.
3. Delete the existing password.
4. Click **OK** and save the file.

Practical: Password Protect a Microsoft Excel Workbook

Steps

1. Open **Microsoft Excel**.
2. Create a workbook and enter some data.
3. Click **File**.
4. Select **Info**.
5. Click **Protect Workbook**.
6. Choose **Encrypt with Password**.
7. Enter a password.
8. Re-enter the password to confirm.
9. Save the workbook.

Test

1. Close the workbook.
2. Open it again.
3. Excel will ask for the password before opening the file.

Remove Password

1. Open the workbook using the password.
2. Go to **File → Info → Protect Workbook → Encrypt with Password**.
3. Remove the existing password.
4. Click **OK** and save the workbook.

Extra Questions

1. What is the full form of SMPS?
2. Why is SMPS required in a computer?
3. What is the full form of UPS?
4. What is the purpose of UPS?
5. What is the difference between AC and DC?
6. Which connector powers the motherboard?
7. Which connector powers the CPU?
8. What is the use of a SATA power connector?
9. What happens if there is no UPS during a power failure?
10. Which command shows battery information in Windows?

Conclusion

- **SMPS** converts AC power into usable DC power for computer components.
- **UPS** provides backup power during electricity failures.
- Students should be able to identify major power cables and connectors and understand their functions.